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Body Language in Sport

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The importance of nonverbal behavior (NVB) in communication is often highlighted with Paul Watzlawick's aphorism "one cannot not communicate" (Watzlawick et al., 1967, p. 51) as people are constantly sending out nonverbal signals that convey some kind of message to observers. While nonverbal communication is often described colloquially as *body language* it can generally be defined as any communicative act not expressed in words. NVB is considered a subprocess of nonverbal communication. It encompasses a wide range of behaviors as all movements can be considered expressive to some degree (Wiener et al., 1972) including facial, vocal, and postural expressions, as well as touch, proxemics, and gaze. However, other aspects of NVB barely classify as behaviors, for example, physical attractiveness, facial morphology, hair style, clothing, jewelry, etc. (e.g., Hess, 2016). Studies suggest that the vast majority of information exchanged between individuals is conveyed nonverbally, although estimates of the exact percentages are believed to range between 65–95% (Matsumoto, Frank, & Hwang, 2013 for a review).

Charles Darwin (1872/1998) is often credited for launching the scientific study of NVB with his seminal book *On the Expressions of the Emotions in Man and Animal*. Darwin's basic argument was that certain internal states, such as emotions, are universally expressed, which he considered an example of an evolved class of behaviors that serve adaptive functions for individuals and species. Darwin tried to answer the question why facial expressions of emotions appear the way they do: for example, why do we wrinkle our nose when we are disgusted or bare our teeth and narrow our eyes when we are angry? His assumption was that these expressions are vestiges of serviceable associated habits (i.e., behaviors that used to have specific adaptive functions). For example, baring the teeth is a prerequisite for an assault amongst species that attack with their teeth; or wrinkling the nose reduced the inhalation of harmful odors. According to Darwin, the formerly functional expressions continued to persist in humans, although they no longer served their original purpose. Or stated differently, the expressions emancipated themselves from their original biological functions (Tinbergen, 1952). In the course of evolutionary history, expressive behaviors acquired communicative value as they provided others with external evidence of an individual's internal state.

Since Darwin's (1872/1998) seminal publication, NVB research has been a focal topic in multiple disciplines. Hence, it is not surprising that a large body of literature has accumulated, but surprisingly the field of sport psychology has been widely unaffected by this research. An exhaustive review of this literature is beyond the scope of this chapter. Therefore, we focus on expressive features of NVB that are associated with certain internal states and (potentially) influence personal and interpersonal outcomes in the broader context of sport. In this endeavor, we first review relevant theoretical work on NVB, the regulation of NVB, how NVB influences impression formation, and exerts its impact on other people. Subsequently, we review and classify research on NVB and impression formation that has been conducted in the context of sports. In the next section, we attempt to integrate existing theoretical and empirical knowledge on NVB to inform evidence-based practice on how athletes, coaches, and sport officials can deliberately use NVB for self-presentational purposes (i.e., impression management).

What Information Does NVB Convey?

Following from the work of Darwin (1872/1998), NVB is most often assumed to convey information about emotion. These emotional expressions inform an individual

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that something important is happening within another person and prepares them to deal quickly with an important event in a way that has been adaptive in the past (Ekman, 1992). The past refers to both individuals' past and what has been adaptive in the history of the species (Ekman & Cordaro, 2011). However, it is important to note that NVB can convey many other kinds of information, such as information relevant to opinions, values, personality dispositions, psychopathologies, physical states such as fatigue, and cognitive states such as comprehension or interest (Fridlund, 1994). Hence, the dominant theoretical notion that NVB—particularly faces, but also vocal, postural expressions, touch, proxemics, and gaze—directly express emotions has come under increasing scrutiny since the 1990s (Fridlund, 1994; Russell, Bachorowski, & Fernandez-Dols, 2003; Russell & Fernandez-Dols, 1997). According to this view, NVB is not necessarily a medium that broadcasts private affective information but should rather be considered a tool for communicating social motives. Or stated differently, NVB is assumed to communicate how people are likely to act, rather than how they are currently feeling. For the purpose of the present review on NVB in sport we do not go into this debate (cf. Parkinson, 2005 for a review) but rather circumvent it by settling on the diplomatic consensus that NVB can convey accurate information about internal states of a person: "Indeed, many emotions are precisely forms of social motives, and many social motives are emotional" (Parkinson 2005, p. 301).

Hence, NVB informs others about a person's current state and how he or she is likely to behave in the proximate future. In social species, this can usually be considered useful information to the one receiving the information and also to the one showing the expression as it helps to organize group life. Organization in groups brings adaptive benefits to animals, but requires efficient communication amongst the individual group members (Dunbar, 1993). In this respect, the communication of internal states is assumed to have ecological utility, as displaying internal states and perceiving and understanding these signaled states enables group members to behave adaptively (McArthur & Baron, 1983). As a result, people have become very proficient at nonverbally displaying internal states and observers can draw accurate inferences of these nonverbal displays (Ambady et al. 2000; Ambady & Rosenthal 1992).

According to *social synapse theory* (Cozolino, 2006) nonverbal communication can be regarded as analogous to the neurochemical communication between synapses in the brain. The social synapse can be regarded as the space through which humans are linked together into larger units such as families, groups, and, indeed, the entire human race, as it conveys the signals that people send out. In turn, these signals are received by our senses and sent to our brains, where they generate electrochemical changes that again create new thoughts and behaviors that are finally transmitted back to the social synapse. Cozolino (2006) goes on to propose that humans efficiently coordinate their social lives by reliably interpreting and acting upon the signals they receive via the social synapse:

> it appears that social communication has been chosen by natural selection to be of greater survival value than disguising our intentions and feelings, so much so that we even have ways of unintentionally "outing" ourselves to others. (p. 24)

In this regard, Dunbar (1993) made the extraordinary discovery of a strong relationship between neocortex size and group-size of social animals. Dunbar argues that the organization in larger groups brings benefits to individuals. However, this requires increasing computational power (larger neocortex) for coordinating the increasing number of relationships, which involves interpreting and acting upon NVB sent out by different individuals into the social synapse (e.g., reciprocal altruism, detecting deception, and coalition formation). Primates with the largest neocortex sizes (i.e. humans) can exploit the adaptive benefits of large groups as they have evolved the necessary computational power that is considered a prerequisite for living in complex societies. This evolved capacity enables humans, amongst other things, to automatically display NVB and to automatically interpret and adequately respond to these NVB (Zebrowitz & Collins, 1997).

Although, it seems like a trivial thought today that NVB can convey accurate information about internal states of a person and that this NVB is understood by untrained, ordinary people viewing this behavior, this knowledge was only established about 60 years ago, most prominently by Paul Ekman (2016):

> As I edit what I have written about what I did nearly sixty years ago, I am still astonished that I had to prove that NVBs are meaningful. It was so obvious; it was soon taken for granted. But then it flew in the face of academic blindness or prejudice about this behavioral domain. And that pleased me no end! I haven't regarded these findings as one of my major scientific contributions, although considering it now, it was a landmark. (p. 7)

Although a large body of theoretical and empirical work suggests that NVB often conveys accurate information about internal states of people, it is important to note that people can also "fake" certain nonverbal expressions and thereby deceive observers (a point we will return to in more detail throughout this chapter). This is especially important in the context of sports as athletes permanently try to deceive their opponents about their behavioral intentions by using specific NVB, like body movements or postures, called fakes (Güldenpenning, Kunde, & Weigelt, 2017; Kunde, Skirde, & Weigelt, 2011).

The Regulation of NVB

Since Darwin's (1872/1998) book on the expression of emotions, many scholars have been interested in the spontaneous and automatic expressions of internal states. However, when considering NVB in sports, it is important to note that NVB is not always automatic and typically not completely outside of intentional regulation. Usually athletes are exerting some control over their NVB, particularly when interacting with team mates and opponents. Even if this attempted control is not always conscious and not always successful, it is pervasive (DePaulo, 1992). In this respect, Erving Goffman (1959, 1963, 1971) was one of the first to explicitly state that people can purposely control their NVB to convey particular impressions. Although Goffman's analyses were based on casual observations and anecdotal evidence, there is a vast amount of experimental evidence in the literature showing that NVB can also be used for self-presentation purposes (see DePaulo, 1992 for an early review). Today, there is general consensus that NVB is under both conscious, deliberate control and unconscious, autonomous control (e.g., see Matsumoto, Frank, & Hwang, 2013 for a review). Hence, NVB is conceptualized to systematically vary along a continuum of controllability (e.g., Ekman & Friesen, 1969; Rosenthal & DePaulo, 1979a, 1979b). For example, two distinct neural pathways have been identified for controlling facial expressions: the pyramidal tract deriving from the cortical motor strip drives voluntary facial actions, whereas the extrapyramidal tract originating in subcortical areas drives involuntary expressions (Rinn, 1984). In this respect, Darwin (Darwin, 1872/1998) asserted that muscles that are difficult to voluntarily activate might escape efforts to inhibit or mask expressions, revealing true internal states:

> When movements, associated through habit with certain states of the mind, are partially repressed by the will, the strictly involuntary muscles, as well as those which are least under the separate control of the will, are liable still to act; and their action is often highly expressive. (p. 54)

Self-presentation refers to a person trying to regulate their behavior in a deliberate fashion in order to create a particular impression on others (Jones & Pittman, 1982), of communicating a particular image of oneself to others (Baumeister, 1982), or of showing oneself to observers to be a particular kind of person (Schlenker & Weigold, 1989). One of the most important questions on NVB in sports is the question of when NVB can and cannot be deliberately produced and when it might leak information that athletes or officials would prefer to deliberately hide. Typically, people assume that desired nonverbal expressions can be produced whenever and without any problems (DePaulo, 1992). However, this is often more difficult than one assumes.

For example, if professional sport referees want to convey an impression of competence when communicating a decision to an athlete, they might instruct themselves to stand erect, hold their head high, directly look at the athlete, and speak calmly in an unwavering voice. Unfortunately for referees, research has shown this might not be as easily done during competition as desired (Furley & Schweizer, 2016b), and that referees' valued goal of appearing competent might in fact undermine their ability to produce the desired NVB. Further, they might sometimes be "trying too hard" which again might be perceived negatively. In a multi-experiment study, the NVB of professional referees was rated as significantly less confident by observers when communicating relatively ambiguous decisions in comparison to relatively unambiguous decisions. The ambiguity of the decision was determined prior to the study by having expert soccer referees decide on a large selection of soccer situations. In the main study, observers had to judge how confident a referee appeared when communicating a decision in a short video clip without seeing the situation that led up to the decision (Furley & Schweizer, 2016b). One explanation how these findings might have occurred is that the referees might be so sure that they already present themselves as desired, although observers do not perceive them that way. As referees are not aware of how their NVB is perceived, they might not invest necessary effort in changing their NVB when communicating decisions. Interestingly, NVB is less accessible to actors than to observers: "In most ways, people know more about themselves, and they know it more directly, than others could ever know. Awareness of NVBs is an important exception to this rule" (DePaulo, 1992, p. 206). We return to the deliberate use and modification of NVB in sports context in the practical application section of this chapter.

NVB and Impression Formation

Observers of NVB are inclined to form some sort of impression based (to some extent) on the NVB they are observing (cf. Kleck & Strenta, 1980). It is impossible to convey no particular impression nonverbally as people cannot refrain from behaving nonverbally (Argyle, 1972).

From an ecological perspective (McArthur & Baron, 1983; Zebrowitz & Collins, 1997), our cognitive system has become particularly attuned to nonverbal cues that are of general adaptive relevance, such as basic emotion cues (Ekman, 1992) or nonverbal cues regulating group lifelike status and dominance cues (Anderson, Hildreth, & Howland, 2015; Schmid Mast & Hall, 2004). Efficient perception of such NVB is most likely not limited to its utility for the survival of species (e.g., a group member signaling a potential threat), but also serves adaptive functions at the level of individual goal attainment, such as determining who should be avoided in a confrontational situation (McArthur & Baron, 1983; Zebrowitz & Collins, 1997). Hence, ecological theories of person perception argue that NVBs that are most important for reproductive fitness are directly perceived, that is, that the information processing system of humans is hard-wired to pick up this kind of information without needing any additional contribution of higher order cognitive processes. For example, threat related signals are assumed to be particularly important for reproductive fitness as perceiving these signals and responding adaptively decreases the chances of a life-threatening attack and, in turn, increases the chances of reaching reproductive maturity (Furley, 2019).

The practical implications of ecological theories seem to be substantial at first sight. People who can "fake" certain nonverbal expressions that are believed to be of adaptive importance (e.g., dominance or high status) can tap into hard-wired perceptual structures—brain networks that are organized in advance of experience, albeit modifiable by experience (Marcus, 2004)—and thereby influence their interaction partners (see Buck's 1984 discussion of "voluntary expression initiation"). However, more recent theorizing (Freeman & Ambady, 2011) suggest that this might be an oversimplification and that impression formation in real life is better regarded as constant interaction among high-level categories, stereotypes, and the low-level processing of facial and bodily cues (i.e., direct perception proposed by ecological theorizing) which dynamically unfold over time. This *dynamic interactive theory of person construal* (Freeman & Ambady, 2011) integrates schema-driven (e.g., Fiske & Taylor, 1991) and data-driven approaches of person perception (McArthur & Baron, 1983; Zebrowitz & Collins, 1997) and attempts to provide a comprehensive account of person perception in situations representative of everyday life. For example, a boxer might be intimidated by an opponent's signals of dominance and physical formidability via direct perception (data-driven), but higher cognitive processes (schema-driven) are theorized to dynamically influence the resulting impression by further taking, for example, personality characteristics, the past success of the boxer, and personal success against him into account.

Although contemporary theorizing (Freeman & Ambady, 2011) situates the role of NVB in the broader cognitive process of person perception and impression formation and thereby mitigates NVB's potential impact in real life social encounters such as sports, there remains reason to believe that NVB can be of impact in real life sporting contexts. In this regard, Nobel laureate Daniel Kahneman (2011) argued that people typically do not acknowledge that they might be missing important information when forming impressions of others. Instead, they tend to treat the limited information available as if it were all there is to know, which Kahneman explains with reference to his "WYSIATI rule" ("What you see is all there is"). Rather than drawing upon or seeking further information before forming an impression of another person, Kahneman (2011) proposed that:

> You build the best possible story from the information available to you, and if it is a good story, you believe it. […] Our comforting conviction that the world makes sense rests on a secure foundation: our almost unlimited ability to ignore our ignorance. (p. 201)

This would suggest that NVB can indeed have an impact in social interactions, particularly in instances in which we do not have much other information besides NVB. Given that these kinds of situations are replete in modern life and in sports as we constantly interact with people for whom we barely have more information than their appearance and NVB, it seems likely that NVB often plays an important role in the impression formation process.

In this regard, a large body of evidence utilizing the *thin-slices paradigm* shows that people can draw accurate inferences from the observation of NVB, even if only very limited information is available. Within the *thin-slices paradigm* observers typically watch short video recordings or images of target people (i.e. the sender of NVB) and are asked to form judgments, infer personality characteristics or internal states, or predict a likely outcome of a social interaction. These observer judgments or estimates are subsequently compared with reliable measurements of the variables in question (see Ambady & Rosenthal, 1992; Ambady et al., 2000 for an overview). Empirical research has demonstrated that participants (i.e. observers) are typically very accurate in this paradigm. Important findings show that observer judgments correspond to a high degree with personality characteristics and dispositions of individuals displaying NVB (Borkenau et al., 2004; Todorov et al., 2005; Willis & Todorov, 2006). Further, observers have been shown to accurately infer mental states of perceived individuals (e.g., Baron-Cohen et al., 1996), what the outcome of a perceived conversation is going to be (Curhan & Pentland, 2007), how student evaluations of a perceived teacher/professor are going to be (Ambady & Gray, 2002), how successful CEOs run a company (Rule & Ambady, 2008), and how successful salespeople are (Ambady et al., 2006). In line with the reviewed theoretical accounts of the effects of NVB on impression formation, accurate judgments of NVB that are especially important to survival (e.g., dominance, fear, disgust) can be made based on very short recordings of a target NVB, whereas longer exposure times are needed when making more complex judgments such as personality variables (Carney et al., 2007).

Although clichés like "one should not judge a book by its cover" are often utilized in popular culture, the outlined evidence highlights that observers can draw accurate inferences about internal states (and more enduring personality characteristics) based on very short observations of NVB—especially NVB that has ecological utility like dominance. However, it is important to note that further cognitive processing, including the context of the interaction, might mitigate the impact of NVB in social interactions in sports. For example, a soccer goalkeeper might observe a highly assertive penalty preparation of a penalty-taker that might initially trigger a favorable impression of the penalty-taker and lower the confidence of the goalkeeper in saving the upcoming penalty (Furley, Dicks, & Memmert, 2012). However, the goalkeeper may then remember that he has a positive history of saving kicks against the respective player. In this case, the perception of the assertive NVB, the stored memory of interacting with that particular penalty-taker, and the context of the current interaction are likely to be integrated in a dynamic fashion and influence the overall impression (Freeman & Ambady, 2011).

How Does the Perception of NVB Influence Observers?

The display and observation of certain NVBs does not only affect impression formation, but has further been shown to affect a variety of subsequent interpersonal outcomes, including cognition, emotion, behavior, and performance (e.g., Argyle, 1972; Freeman & Ambady, 2011; Fridlund, 1994; Matsumoto et al., 2013; Warr & Knapper, 1968 for reviews). Although, the reviewed ecological theories of person perception (McArthur & Baron, 1983; Zebrowitz & Collins, 1997) predict that the perception of certain NVBs that are of high ecological utility (e.g., threat displays) might determine behavior directly (without further higher order cognitive involvement), the influence of perceiving NVB on behavior is usually assumed to occur via cognitive and emotional processing.

Early theorizing by Warr and Knapper (1968) assumed that the perception of NVB in another person led to affective, attributive, and expectancy responses which in turn influenced behavior toward that person. While these mechanisms are still integral parts in more contemporary theorizing on the effects of perceiving certain NVB in other people (Matsumoto et al., 2013 for an overview), these are not the only mechanisms of how NVB can impact on behavior and performance in sports. On a very general level, van Kleef (2009) proposed the e*motions as social information model (EASI-model)* to better understand how (emotional) NVB may exert interpersonal effects. Central to the model is the assumption that NVB affects observers via two routes: inferential processes and/or affective reactions. Inferential processes describe how an observer of NVB is able to infer certain information about internal states (e.g., feelings, attitudes, relational orientations) of other people. Observers use this information to better evaluate the situation and this information helps them to deliberate on an adequate behavioral response. For example, when one is observing a display of pride, one may conclude that this individual has achieved something important (inference), and should be treated in accordance with this achievement (e.g., respectfully, Parkinson, 1996). On the other hand, the observed expressions can directly elicit affective reactions within the observer. One type of affective reaction occurs via the process of emotional contagion whereby individuals catch the expresser's emotions through his or her facial expressions, bodily movements and postures, or vocalizations (Hatfield, Cacioppo, & Rapson, 1993). Most of the interpersonal effects of NVB in the context of sports can be explained via the two general mechanisms of inferential and/or affective processing.

NVB in Sports

The reviewed literature shows that NVB changes as a consequence of situational variables, either because a person automatically shows a nonverbal reaction to triggering internal and external circumstances (e.g., as is theorized to occur for some basic emotions expressed in the face), or because a person deliberately wants to convey certain information nonverbally to observers in a given situation. The display and observation of certain NVBs has been shown to have a variety of effects on subsequent interpersonal outcomes, including cognition, emotion, and behavior (e.g., Argyle, 1972; Matsumoto et al., 2013 for reviews). In the next section we review the literature on NVB in the context of sports to address two main questions. First, does NVB change as a result of sports performance? Second, does sports performance change as a consequence of NVB? Furthermore, there are several questions related to these overall questions concerning mediators and moderators that we attempt to address alongside the two main questions before we conclude this section by highlighting necessary and fruitful avenues for future investigation.

Research on NVB in Sports

Existing studies on NVB in sports differ regarding type of NVB they investigate (pre-, during-, or post performance), theoretical background (evolutionary theorizing, schema theory, or emotion expression), methodological approach (experimental or observational; quantitative or qualitative) and stimulus material (naturalistic vs. artificial), just to name a few potentially meaningful differences between existing studies. Instead of using one (or all) of these aspects to systematically distinguish between studies, we focus on NVB as either a consequence or a predictor of sports performance. NVB as a consequence of sports performance means that athletes experience some sort of success or failure and subsequently, their NVB changes. NVB as a predictor of sports performance means that NVB changes and subsequently, sports performance changes. Furthermore, an ensuing research question relates to mediating variables: which variables mediate the NVB-performance relationship, if at all? Finally, which NVBs play a role for the different research questions depicted above, both as expressions of different internal states and as cues for their perceptions?

NVB as a Consequence of Sports Performance

Several studies suggest that NVB differs following successful or unsuccessful sports performances (e.g., Aviezer, Trope, & Todorov, 2012; Moesch, Kenttä, Bäckström, & Mattsson, 2015; Moesch, Kenttä, Bäckström, & Mattsson, 2016; Furley & Schweizer, 2014a; Furley & Schweizer, 2016a; Hwang & Matsumoto, 2014; Ryan, Furley, & Mulhall, 2016; Tracy & Matsumoto, 2008; Whittaker-Bleuler, 1980). Furthermore, studies using the *thin-slices* paradigm suggest that perceivers can distinguish between NVB shown after successful or unsuccessful performances (Furley & Schweizer, 2014a; Furley & Schweizer, 2016a). Differences in NVB after success or failure seem to be a rather general phenomenon, not confined to specific sports or competitions. Likewise, the ability to distinguish between NVB after success or failure seems to be a general one, not restricted to age, gender, or expertise (Furley & Schweizer, 2014a). There is evidence for nonverbal changes after success or failure from several kinds of sports, both individual and team (handball, basketball, table tennis, judo, badminton), across a range of different competitions, both international (Olympic and Paralympic Games, World Cups) and national (e.g., NBA, highest national leagues in handball and basketball). Initial evidence suggests that nonverbal emotional reactions towards success or failure tend to "leak," meaning that athletes cannot entirely control their nonverbal reactions, even if they try (Furley & Schweizer, 2016b). Research following this tradition typically used naturalistic stimulus material, such as video recordings of behavior in real sports competitions. For example, participants watch brief videos from basketball matches that show one or several players during breaks in the game (e.g., time-outs, Furley & Schweizer, 2014a). However, videos do not show obvious indicators of success or failure (e.g., raising the fists or hiding the face). For every video, participants are asked to estimate on a continuous scale whether the depicted players are trailing or leading. This procedure allows investigation of whether participants are able to distinguish between leading and trailing athletes based on their NVB by comparing participants' estimates to the real score in every video clip.

Success or failure has been operationalized in different manners, from winning or losing a whole match (e.g., Tracy & Matsumoto, 2008); to leading or trailing as indicated by the current score in an ongoing competition (e.g., Furley & Schweizer, 2014a); to behaviors shown directly after a single successful or unsuccessful attempt (e.g., a shot at the goal; Moesch, Kenttä et al., 2015; Moesch et al., 2016). Some studies have focused on NVBs that are clear indicators of success or failure (e.g., raising the fists or high-fiving; Moesch, Kenttä et al., 2015; Moesch et al., 2016), whereas others excluded these clear indicators because they focused on less intentional expressions of internal states (e.g., standing in a more or less expansive manner; Furley & Schweizer, 2014a, 2016b).

Overall, findings from studies tend to be in line with evolutionary accounts of NVB (Furley, 2019) which assume that (1) as a result of winning or losing in antagonistic encounters humans change their NVB in order to communicate dominance and pride (i.e., high status) or submission and shame (i.e., low status), and that (2) other humans are able to detect and interpret these changes (Furley & Schweizer, 2014a, 2016a; Tracy & Matsumoto, 2008). These findings have been interpreted as being the result of adaptations that evolved to solve specific adaptive problems in our ancestral past: sending signals when losing an antagonistic encounter increased the chances of avoiding further potentially life-threatening attacks. Likewise, sending victorious signals helped to save valuable resources by communicating high status and superiority over the opponent, and further signal supremacy to potential mates. This assumption is supported by results showing that leading athletes are perceived to be higher on dimensions related to social status (i.e., dominance, pride, confidence) than trailing athletes (Furley & Schweizer, 2016a). Likewise, athletes display pride following success and shame following failure (Tracy & Matsumoto, 2008). Studies suggest that both body-related and face-related cues signal past performance (for peak-intensity emotional expressions, however, the face alone does not seem to be diagnostic) (Aviezer et al., 2012), and perceivers can decode them even from briefly presented stills (Furley & Schweizer, 2016a). Currently there is only little evidence showing which nonverbal reactions exactly occur as a consequence of success or failure and which nonverbal cues perceivers in turn use.

NVB as a Predictor of Sports Performance

Few studies investigated a direct link from NVB to sports performance. Evidence suggests that in handball, NVB and prior performance interact when predicting future performance (Moesch et al., 2016). Positive subsequent team performance was predicted by a high degree of touch when playing well and a low degree of touch when playing poorly. In turn, negative subsequent team performance was predicted by a low degree of touch when playing well and a high degree of touch when playing poorly. In basketball, touching behavior early in the season predicted both individual and team performance later in the season, even when potential confounding variables were controlled for (Kraus, Huang, & Keltner, 2010). This study focused on 12 intentional forms of touch, such as fist bumps, high fives, head slaps, or hugs. One explanation for the relationship between touch and performance might be that touch either promotes or is an indicator of cooperation which, in turn, positively affects performance. In penalty shoot-outs, certain celebratory behaviors shown by successful penalty-takers predict the success of the next penalty taken by a member of the opposing team and the likelihood of the celebrating penalty-taker being in the winning team (Moll, Jordet, & Pepping, 2010). For example, when players raised both arms above their heads after a successful penalty, the next shooter was more likely to fail and the celebrating player's team was more likely to win. Further research suggests that higher pressure during a penalty shootout is related to avoidance NVB such as looking away from the keeper and speeding through the preparation before the kick (Jordet, 2009a, 2009b; Jordet & Hartmann, 2008). In this line of research, looking away from the keeper and speeding through the preparation before the kick are considered to be avoidance behaviors as they are assumed to represent a tendency to get out of an uncomfortable situation as quickly as possible. However, among these avoidance NVBs only, preparation times have been shown to directly predict penalty performance (Jordet, 2009a; Jordet & Hartmann, 2008).

Mediators of the Relationship between NVB and Performance

Many studies assume that NVB does not necessarily directly influence sports performance itself, but that NVB influences variables that may indirectly influence sports performance. Studies in this research tradition may follow different approaches that differ both theoretically and methodologically. In the first approach, researchers experimentally manipulate NVB and investigate into the effects these manipulations have on impressions and expectancies formed by teammates and opponents. Theoretically, this line of research is derived from theories of impression formation (Freeman & Ambady, 2011; Warr & Knapper, 1968). Methodologically, it focuses on manipulations of NVB that are usually displayed as video clips. In the second approach, researchers conduct interview or questionnaire studies in order to investigate into athletes' perceptions of the impact NVB has on their confidence. Theoretically, this second line of research focuses on theories of collective efficacy and team outcome confidence (Bandura, 1997). Methodologically, it focuses on questionnaire and interview studies. Due to employing mostly experimental designs and controlled manipulations of NVB, the first approach allows for deriving causal conclusions on the impact of NVB on impression formation. In turn, due to investigating into athletes' perceptions on NVB, the second approach allows for deriving conclusions on the perceived importance of NVB in the field, mainly as a factor influencing confidence.

NVB and Impression Formation: Experimental Studies

In the first approach, researchers assume that athletes' NVB may affect the impressions and expectancies both their teammates and opponents form (Buscombe, Greenlees, Holder, Thelwell, & Rimmer, 2006; Furley & Dicks, 2012; Furley, Dicks, & Memmert, 2012; Furley, Dicks, Stendtke, & Memmert, 2012; Furley, Moll, & Memmert, 2015; Furley & Schweizer, 2014b; Greenlees, Bradley, Holder, & Thelwell, 2005; Greenlees, Buscombe, Thelwell, Holder, & Rimmer, 2005; Rejeski & Lowe, 1980). Impressions and expectancies in turn may (again directly or via additional mediating variables) influence performance. This approach rests on theories of impression formation, person perception, and schema-driven processing (e.g., Fiske & Taylor, 1991; Warr & Knapper, 1968). The perception of NVB in early stages of a social interaction between athletes is supposed to activate a person schema (e.g., "a good athlete", or "a powerful shooter"), which in turn influences further perceptions and information processing.

Research following this tradition mostly uses experimentally manipulated stimulus material, such as videotapes of actors who were instructed how to portray positive or negative NVB (but see Furley & Schweizer, 2014b, for a study using naturalistic stimulus material). Actors are instructed to display positive NVB by maintaining an erect posture (shoulders back and chest out), keeping their head up, and looking directly into the camera. In turn, they are instructed to display negative NVB by maintaining a hunched posture, keeping the head down, and not looking directly into the camera (see for example Buscombe et al., 2006). This manipulation is not theoretically informed by concepts such as dominance and submission or pride and shame, but by guidelines for positive body language from applied sport psychology (Weinberg, 1988).

Still, it is easy to see how these guidelines overlap with both the description of dominant and submissive NVB and the description of pride and shame, although some differences remain (e.g., expansion and constriction do not play a central role). Some studies, however, use a manipulation of NVB that directly refers to dominance and submission or pride and shame by showing the biological motion information of these NVBs using point light stimuli (e.g., Furley, Dicks, & Memmert, 2012; Furley, et al., 2015).

Studies suggest that athletes' NVB indeed influences opponents' anticipated emotions, outcome expectations, impressions of the opponent, and episodic as well as dispositional judgments of the opponent (e. g., Buscombe et al., 2006; Greenlees, Bradley et al., 2005; Greenlees, Buscombe et al., 2005). When opponents display positive NVB, athletes (1) believe they are less likely to perform successfully against them, (2) perceive them as more favorable along several stable characteristics such as assertiveness, aggressiveness, and competitiveness and (3) judge their current states (such as confidence, focus, and tension) more positively. For example, in a study by Greenlees, Bradley, and colleagues (2005), participants who were experienced table-tennis players viewed videos of actors during a warm-up before a table-tennis match. Actors were instructed to either display positive or negative NVB. Participants were more confident to defeat actors showing negative NVB and they rated actors showing negative NVB less positively (e.g., less assertive, less competitive, less fit, less focused, more on edge). Similar results were obtained in studies conducted with tennis players (Buscombe et al., 2006; Greenlees, Buscombe, et al., 2005).

Furthermore, studies suggest that athletes' NVB not only influence general impressions regarding their mental state or their personal characteristics, but also sport-specific judgments of their abilities that may be directly relevant for performance (Furley, Dicks, & Memmert, 2012; Furley, Dicks, et al., 2012). For example, baseball players expect pitches to be of lower quality when pitchers show negative NVB (Furley & Dicks, 2012), just like goalkeepers expect a penalty to be more powerful when a shooter shows positive NVB (Furley, Dicks, et al., 2012). This expectation has a direct effect on behavior: goalkeepers initiate their movements earlier in response to a penalty taken by a shooter with positive NVB than in response to a penalty taken by a shooter with negative NVB. Researchers have tried to identify interactions between NVB and cues that are acquired via cultural learning (e.g., clothing style), however, so far results remain inconclusive (Greenlees, Buscombe, et al., 2005; Furley, Dicks, & Memmert, 2012). Again, these findings do not seem to be confined to specific sports, as there is evidence for the effect of NVB on impressions and expectancies from different sports, both team and individual (e.g., soccer, baseball, tennis, table-tennis). Furthermore, athletes rate NVB to be among the most important information (next to coaching experience, clarity of voice and success rate) used when forming impressions of coaches (Manley, Greenlees, Graydon, Thelwell, Filby, & Smith, 2008).

In studies on the impact of NVB on the impressions formed of penalty-takers, a more penalty-specific manipulation of NVB has been employed, the manipulation of so-called hastening and hiding behavior (Furley, Dicks, et al., 2012). Hastening and hiding is defined by the time penalty-takers take during preparation of the penalty (i.e., the time between the referee blowing the whistle and the shooter beginning the run-up) and by the level of eye contact they maintain with the goalkeeper (i.e., the shooter looking into the eyes of the goalkeeper more often and longer). The rationale underlying this manipulation is that evidence from real penalties suggests that longer preparation times and higher levels of eye contact towards the keeper are related to penalty performance (Jordet, 2009a; Jordet & Hartmann, 2008).

NVB and Confidence: Athletes' Perceptions

In the second approach, researchers focus on the idea that athletes' NVB may affect the confidence both their teammates and opponents experience. Contrary to the approach depicted above, evidence for the role of NVB for variables such as self-efficacy and team outcome confidence mainly comes from interview and questionnaire studies. Both athletes and coaches regard NVB to be one of the most important sources for collective efficacy and team outcome confidence (Fransen, Vanbeselaere, de Cuyper, Vande Broek, & Boen, 2015; Fransen, Vanbeselaere, Exadaktylos, Vande Broek, de Cuyper, Berckmans, et al., 2012; Ronglan, 2007). Results suggest that athletes' confidence depends on their teammates' NVB: athletes are more confident to win when their teammates show positive NVB and they are less confident to win when their teammates show negative NVB. In addition to its relevance for team efficacy, coaches regard NVB to be both a characteristic and a trigger for positive and negative psychological momentum in sports (Moesch & Apitzsch, 2012). Again, the role of NVB for (team) confidence is not confined to specific sports but has been shown in various sports (e.g., handball, volleyball, basketball, and soccer).

Taken together, evidence from experiments (employing both artificial and naturalistic stimulus material), observational studies and interview or questionnaire studies suggests that athletes' NVB influences their teammates' and opponents' emotions, impressions, expectancies, and confidence. These variables may in turn either directly affect athletes' performance or they may affect further mediating variables. Particularly the finding that NVB affects confidence can be considered indirect evidence that NVB can indeed affect sports performance, as confidence has been shown to influence sports performance (Feltz & Lirgg, 1998; Fransen, Decroos, Vanbeselaere, Vande Broek, De Cuyper, Vanroy, & Boen, 2015; Fransen, Mertens, Feltz, & Boen, 2017; Greenlees, Nunn, Graydon, & Maynard, 1999; Myers, Feltz, & Short, 2004; Myers, Payment, & Feltz, 2004).

Identification of Specific NVBs in the Field

The research approaches depicted above either focus on NVB as a consequence of success or failure or on the effects of NVB on impressions or confidence; however, they do not primarily attempt to identify which nonverbal behaviors specifically exert the depicted effects. In an additional line of research, new methods have been developed with the goal of identifying specific NVBs. Some studies have attempted to identify which NVBs occur during real sports competitions by using elaborate coding schemes (Hwang & Matsumoto, 2014; Kraus et al., 2010; Matsumoto & Hwang, 2012; Moesch, Kenttä, & Mattsson, 2015, Moesch et al., 2015, 2016; Moll et al., 2010; Kneidinger, Maple, & Tross, 2001; Tracy & Matsumoto, 2008). Some of these coding schemes are sport specific, but others are not. For example, the Handball Post-Shot Behavior Coding Scheme (H-PSB-CS; Moesch, Kenttä, & Mattsson, 2015) was developed in order to code post-shot behaviors by female handball players, whereas Tracy and Matsumoto (2008) have suggested a more general system for coding pride and shame. This system was later adapted in order to code "triumph" shown by winners but not by losers (Hwang & Matsumoto, 2014; Matsumoto & Hwang, 2012). Studies using coding schemes provide detailed analyses into NVBs displayed during sports competitions. For example, studies using the H-PBS-CS highlight the role of specific gestures (e.g., one fist up, two fists up, thumbs up) and touch behaviors (e.g., low five, high five, touch shoulders) (Moesch, Kenttä, et al., 2015; Moesch, Kenttä, & Mattsson, 2015; Moesch et al., 2016). Other studies highlight gender differences in touching behavior (Kneidinger et al., 2001). When comparing results from different sports such as handball (Moesch, Kenttä et al., 2015; Moesch, Kenttä, & Mattsson, 2015; Moesch et al., 2016), softball or baseball (Kneidinger et al., 2001) and basketball (Kraus et al., 2010), it appears that some NVBs are sport specific, whereas others are not or at least to a lesser extent. Studies employing coding schemes have different theoretical backgrounds, which can in turn influence the primary focus of the coding scheme. For example, some focus on the expression of pride and shame (e.g., Moll et al., 2010; Tracy and Matsumoto, 2008), whereas others strongly focus on touching behaviors (e.g., Kneidinger et al., 2001; Kraus et al., 2010). Owing to differences regarding coding schemes' theoretical background, construction principles, and potential differences between coding situations and sports, it is hard to compare different studies using coding schemes.

Open Research Questions

Causality of Effects

Studies suggest that NVB changes following success or failure (see section on NVB as a consequence of sports performance) and that certain NVBs precede changes in sports performance (see section on NVB as a predictor of sports performance). However, these studies do not show that success or failure directly cause the reported changes in NVB. Neither do they show that NVB causally affects performance. Future researchers might therefore attempt to experimentally manipulate success or failure to look for subsequent effects on NVB. Likewise, they might attempt to experimentally manipulate athletes' NVB and look for subsequent effects on performance.

Effects from NVB on performance might follow different pathways: athletes' NVB might affect (1) their own performance, (2) their teammates' performance, (3) or their opponents' performance. Causal effects from NVB on performance would be particularly important for practitioners (e.g., athletes, coaches, sport psychological consultants). Research on the question whether athletes can deliberately manipulate their NVB to improve their performance (either by directly improving their own performance or by improving teammates' performance or by worsening opponents' performance) needs to be based on a solid theoretical foundation. Whereas theoretical foundations exist for effects on teammates' or opponents' performance (e.g., theorizing on self-efficacy; Bandura, 1997), currently no equally strong theoretical framework exists for effects on athletes' own performance. Although first results from social psychology suggest that deliberately manipulating one's NVB may have several potentially beneficial effects (e.g., Carney, Cuddy, & Yap, 2015; Cuddy, Wilmuth, Yap, & Carney, 2015), some of these results have been disputed (e.g., Cesario & Johnson, 2018; Gronau, Van Erp, Heck, Cesario, Jonas, & Wagenmakers, 2017; Ranehill, Dreber, Johannesson, Leiberg, Sul, & Weber, 2015).

Assuming that sports performance indeed affects NVB and that NVB indeed affects sports performance, it seems feasible that performance and NVB mutually influence each other in an iterative manner, thus leading to some sort of self-reinforcing cycle or loop (for first evidence supporting this notion see Moesch et al., 2016). This hypothetical cycle could start out both with changes in performance and with changes in NVB. Future studies might investigate into multiple such iterations, each one reinforcing another. This research seems fruitful to combine with momentum approaches (Moesch & Apitzsch, 2012; Iso-Ahola & Dotson, 2016) and dynamical system approaches to sports performance (Vilar, Araújo, Davids, & Button, 2012).

Underlying Processes and Moderators

Several studies report effects of NVB on impressions and expectancies (see section on mediators of the relationship between NVB and performance). As many of these studies are based on experimental manipulations, it seems safe to conclude that NVB does indeed influence emotions, cognitions, impressions, and expectancies. However, although most studies in this research tradition assume that effects are mediated via the activation of a person schema, there is only sparse evidence for this assumption. Future research needs to address the question whether effects of NVB on impressions and expectancies are a product of bottom-up (i.e., more data-driven) or top-down (i.e., more schema-driven) processing or of an interaction between both processing modes (Freeman & Ambady, 2011; Furley et al., 2012). Research investigating into this question might be informed by theorizing about different kinds of interactions between more automatic and more deliberate processing (Furley, Schweizer, & Bertrams, 2015).

Furthermore, future research might benefit from investigating into the question how and when nonverbal responses that are to some extent "evolutionarily inherited" are moderated or even overridden by cultural learning. Tracy and Matsumoto (2008) provide evidence for the idea that nonverbal displays of pride and shame after success and failure are at least partly biologically innate by showing that they occur for both sighted and congenitally blind athletes. However, this does not mean that nonverbal reactions are entirely innate and culture-independent (Tracy & Matsumoto, 2008). Additionally, not only the display of nonverbal reactions, but also the ability to correctly encode them might be moderated by psychological variables. For example, theorizing on emotional competence (or emotional intelligence) assumes that individuals differ regarding their ability to identify emotional expressions (Petrides & Furnham, 2003). Likewise, individuals who have a high need for power should be particularly sensitive towards cues associated with power (Donhauser, Rösch, & Schultheiss, 2015). Therefore, it seems feasible that perceivers with high emotional intelligence or a strong power motive are more efficient at decoding these nonverbal cues during sports competitions and might react more strongly towards these cues. Initial results suggest that combining methods from research on NVB in sports and methods from neuroscience may prove beneficial, e.g., in scrutinizing the evolutionary theorizing of NVB occurring as a consequence of success and failure (Furley, Schnuerch, & Gibbons, 2017).

Identifying NVB in Sports

While research has demonstrated that NVB changes as a consequence of performance and that NVB can influence impressions and expectations, little is known about the precise NVBs that occur and change during competitions. In addition, further research has shown that NVB is influenced by other factors than performance in sports competitions as, for example, by whether athletes are playing at home or away (Furley, Schweizer, & Memmert, 2018) or whether they interpret a certain situation as a challenge or a threat (Brimmell, Parker, Furley, & Moore, 2018). Therefore, future research would benefit from developing more elaborate coding schemes for NVB in sports. It is a research question in itself whether these coding schemes need to be highly domain specific like the H-PSB-CS (Moesch, Kenttä, & Mattsson, 2015) or whether the development of general coding schemes is feasible. Evolutionary theorizing suggests that NVB (as expression of internal states) should be highly general, whereas other approaches may highlight sport-specific differences regarding NVB (e.g., touching in team sports vs. raising the fist in individual sports). Therefore, future research might use already existing coding schemes from domains other than sports, like the Facial Action Coding System (FACS) (Ekman & Friesen, 1978) or the Body Action and Posture Coding System (BAP) (Dael, Mortillaro, & Scherer, 2012). Alternatively, future research might use automated facial coding software, like FaceReader (den Uyl & van Kuilenberg, 2005; Noldus, 2014).

Related to the above question, so far results are preliminary regarding the relatively stronger impact of positive or negative NVB and the role of neutral NVB (e.g., Furley & Dicks, 2012). Do success and failure equally affect positive and negative NVBs? Do positive or negative NVBs affect team members' and opponents' impressions and expectancies more strongly? What impact do neutral expressions have? Obviously the open research questions outlined in this section only represent a fraction of potential further research avenues on NVB in sports, but, in our opinion, they entail the most important ones and the ones that follow most directly from the research that has been carried out on NVB in sport so far.

Practical Implications

Throughout this chapter, we have reviewed evidence showing that many instances of NVB are not deliberate and even unintentionally leak information that people want to hide. However, in many situations, people, and especially high-level athletes and officials, deliberately try to present themselves in a particular manner as can be demonstrated by the following quote of a world-class athlete:

> I think even if you're not confident inside, you need to present yourself as confident on the outside because that's half the battle won; firstly with yourself, because if you present yourself as confident then you immediately feel more confident, and also for your opponents, if you look confident then you're obviously a little bit more scary, perhaps they don't feel as confident as you look and might be intimidated by that. (Hays, Thomas, Maynard, & Bawden, 2009, p. 1192)

This quote illustrates a type of impression management (Schlenker, 1980) that not only elite athletes (Hays et al., 2009), but also recreational athletes engage in. As outlined previously, NVB systematically varies along a continuum of controllability (e.g., Ekman & Friesen, 1969; Rosenthal & DePaulo, 1979a, 1979b) with some forms of NVB being displayed automatically without deliberate conscious effort, while other forms of NVB are deliberately used to convey a desired impression. Some forms of NVB (e.g., facial expressions of basic emotions) are innate as they automatically coincide with emotional programs that evolved to prepare humans to behave adaptively (Ekman, 1992), while other NVBs are learned deliberately or implicitly via extensive experience. However, and of importance regarding impression management in sport, certain NVB that used to be deliberately shown for self-presentational purposes can become habitual with extensive practice (Jones & Pittman, 1982), and people can learn to modify, mask, or suppress NVBs that are innate or have become automatic. DePaulo (1992) exemplifies the process of NVBs moving along the controllability continuum via the example of how young girls from traditional families learned to "sit like a lady." This is practiced deliberately for years and eventually conscious monitoring of this particular NVB is no longer necessary and this NVB will be displayed automatically. The terms "deliberate practice" (Ericsson, Krampe, & Tesch-Römer, 1993) or "conscious monitoring" (Fitts & Posner, 1967; Hardy, Mullen, & Jones, 1996) are established concepts in the sport and skill acquisition literature and are deliberately used here to argue that NVB, at least in some instances, can be considered a skill that can be learned and modified for self-presentational purposes in competitive sports.

As we are not aware of any research on the use of NVB for self-presentational purposes in sports, the aim of the following sections is to review evidence in the general literature that can be taken as guidelines for coaches, athletes, sport officials, and researchers of how, and under what conditions, NVB can be used for impression management. An important mechanism governing impression management amongst athletes and sport officials is what Ekman (1972) referred to as display rules which describe cultural norms about what kind of NVBs are appropriate for which person in a certain context. For example, an athlete who is runner up in an Olympic contest is expected not to nonverbally show frustration about losing the gold medal during the award ceremony, but instead acknowledge the success of the winner along with happiness for his success. It is important to note that these kinds of deliberate NVBs do not necessarily have to be deceptive, even if they might be on occasion (e.g., Schlenker, 1980). Most often, impression management involves increasing or decreasing the intensity of the expression of a certain internal state (e.g., athletes wanting to present themselves as less nervous or showing more confidence than they actually have).

The production of certain NVBs to deliberately convey a desired impression in sports sounds simple. Hence, for example, the international soccer association FIFA states that "body language is a tool that the referee uses to help him control the match and show authority and self-control" (FIFA, 2015, laws of the game, p. 84). For this reason, referees are coached on how to deliberately use NVB to communicate decisions confidently in their training programs. Such training programs on NVB typically rest on the premise that anyone physically capable of making certain NVBs can use it—and probably successfully—when trying to convey a certain impression. However, research in the field of sports shows that deliberately trying to convey a certain impression nonverbally might not be as easy as one may think (Furley & Schweizer, 2016b). In this respect DePaulo (1992) systematically summarizes constraints that influence the translation of self-presentational intentions into the actual production of NVBs: "Sometimes the expressions that people would like to enact are ones that they simply cannot produce. Other times, the expressions are ones that they could produce under optimal conditions, but the prevailing circumstances are undermining rather than enabling" (p. 212).

Constraints on the Use of NVB for Impression Management

Translating self-presentational intentions into the actual production of NVB can be problematic at several time points. Numerous factors constrain the nature of the intentions formed, numerous factors disrupt the translation from intention to action, and many factors limit the effectiveness with which people evaluate and modify their NVB (DePaulo, 1992).

Constraints on the Formulation of Intentions to Use NVB

Cultural constraints, situational constraints, and knowledge influence whether the use of certain NVBs is even considered for self-presentation purposes. Every culture has specific norms that inform people what is appropriate for whom in a certain context and what is not. This is also the case for NVB and many instances come to mind how cultural and situational constraints impact on whether people form the intention of using their NVB to convey certain impressions (e.g., a teenager from the United States might nonverbally try to present themselves as cool in front of classmates; while this might not be the case for a teenager going to school in a developing country). Further, people need to have some basic knowledge of the relationship between NVB, internal states, and resulting expression in a particular context to successfully use NVB for impression management.

Constraints on the Translation of Intentions to the Production of NVB

Several factors have been identified that influence how readily people can translate an intention to show a particular NVB into actual NVB (see DePaulo, 1992 for more detail).

Ability, Practice, Experience

Once people have formed an intention and know which NVBs they need to produce to convey a certain impression, they need certain abilities, practice, and experience to enact the NVB. In this respect, it is important to note that many, though not all, NVBs can be improved by practice (DePaulo, 1992) which is why there is, for example, anecdotal evidence that accomplished poker players have become especially talented at regulating their NVBs when they are bluffing (Hayano, 1980).

Constraints on the Controllability of NVB

Physical characteristics, like size, face wrinkles, and skin color impact on the impressions that NVBs convey and can be a source of enormous frustration for people wishing to convey a certain impression as these physical characteristics are hardly controllable. Further, contextual influences like pressure or fatigue can tip the balance between automatic and deliberate control of NVB rendering it harder to control one's NVB for self-presentational purposes in a state of fatigue or depletion (e.g., Matsumoto et al., 2013, for a review). Following from this point, people are limited in the range of impressions they can convincingly claim. For example, a coach with a very high-pitched voice might struggle to convey the impression of a tough, high-powered leader. According to early theorizing, people have individual expressive styles (Allport, 1937; Maslow, 1949) which renders it difficult to successfully regulate some types of NVB as people quickly retreat to the comfort of their habitual expressive style. People also differ in their spontaneous expressiveness when they are not trying to deliberately convey an impression (DePaulo, 1992). These individual differences can be both facilitative and debilitative in producing NVB for impression management.

Motivational and Emotional Constraints

Conveying a desired impression at a certain time in a specific context is typically a strenuous effort, and therefore people sometimes are not motivated to make that effort. Hence, motivation can be considered an important factor at every point in the self-presentational process. Further, given the automatic links between certain basic emotions and NVB, emotions can undermine self-presentational efforts. For example if a tennis player is terribly afraid and nervous to play his first Grand Slam final, he will probably have a hard time to convince observers that he is actually calm and relaxed. Finally, people will be more likely to succeed at conveying an impression nonverbally if they are confident they can do this compared to people who are lacking confidence (DePaulo, Blank, Swaim, & Hairfield, 1992; Schlenker & Leary, 1982).

Constraints on Appraisals and Modification of NVB

DePaulo (1992) argues that online modification of NVB is difficult as people are often not aware of their NVB and therefore struggle to fine-tune their self-presentations. Further, people are often wrong about estimating their ability to control NVB (Zuckerman, Koestner, & Driver, 1981) and err about how they believe other people perceive their NVB (Kenny & DePaulo, 1993).

Becoming Skilled at Conveying Impressions Nonverbally

The outlined constraints suggest that athletes, coaches, and sport officials are likely to struggle occasionally when trying to use their NVB for impression management. However, suggesting that something is difficult should not be taken as discouragement to educate athletes, coaches, and officials in how to use NVB to their advantage. The next section reviews evidence showing that people can (on occasion) overcome the outlined constraints and, subsequently provides guidelines on how to train NVB for impression management.

DePaulo (1992) reviews various studies on people's "posing" abilities in which people are asked to use their NVB to try to convey a particular emotion or internal state that they are not experiencing at the time. This research suggests that most people are fairly skilled posers, especially when using their face and voice. Individual studies also show that untrained people can use their body (Cunningham, 1977) or walking styles (Montepare, Goldstein, & Clausen, 1987) to convey particular internal states they are currently not really experiencing. Similarly, research has shown that people can even actively use their NVB to deceive observers and make them believe they are experiencing something very different from what they are in fact experiencing (e.g., DePaulo & Rosenthal, 1979; DePaulo, Rosenthal, Green, & Rosenkrantz, 1982) at least in short-term interactions (Zuckerman, Amidon, Bishop, & Pomerantz, 1982).

A related stream of research has addressed the complimentary question of whether observers can reliably tell when other people are deliberately regulating their NVB for impression management. As research has shown that NVB can be used to convey an impression that is not genuine, this means on the flip side that observers are sometimes not successful at detecting that they were being deceived by the NVB. However, success rates at using NVB for deception and for recognizing deception vary substantially as a function of the reviewed individual differences and contextual variables (DePaulo, 1992):

> This is perhaps as it should be. It would be neither desirable nor useful to have a social system in which anyone could successfully claim any image at any time. Nor would it do to have a system in which no one could ever succeed at conveying anything other than their genuine feelings. [...] There is much potential, throughout the lifespan, for all interactants to develop and refine their abilities to regulate their own NVBs and to discern others' attempts to do the same. This is part of the richness, flexibility, and intrigue of social life. (pp. 234–235)

Training NVB for Impression Management in Sports

The previous section highlights that deliberately using NVB to one's advantage in the context of sports is complicated by numerous constraints. Hence, it most likely is not sufficient to simply look up NVBs that have been linked to desirable impressions in popular publications on body language (e.g., Pease & Pease, 2005) and try to copy these in certain sport contexts in which one hopes to benefit from posing certain NVBs. Although, there is no solid knowledge base that allows giving evidence-based recommendations for the use of NVB in sports we attempt to outline some general recommendations that will most likely be beneficial in sports.

As an initial step it is important to educate people involved in sports on body language. It is important to know about (1) its assumed evolutionary meaning, (2) that NVB has some innate components associated with hard-wired emotional modules, (3) that it varies along a continuum of controllability, (4) that there are many constraints on the deliberate use of NVB for self-presentational purposes. Given that people are often not aware of their NVB or are mistaken about how they believe other people perceive their NVB, an advisable second step would be to increase awareness of the displayed NVB in certain situations by informing athletes how their body language is perceived and what reactions it might elicit on observing opponents and team mates. This could, for example, be done with video recordings during competitions in combination with team discussions. In this way, it might be possible to identify instances when NVB diverges from intended self-presentation and subsequently attempt to modify NVB in a beneficial way. To this end, it should be possible to shape deliberate practice activities (that take the constraints into account) to move certain desired NVBs to the controllable pole of the continuum and subsequently practice them so that they become automatic (and no longer appear as staged). However, as with every other skill, the deliberate practice activities on NVB have to be thoughtfully developed and practiced over an extended period of time for them to be implemented successfully in competitive sports.

Conclusion

The history of research on NVB in sports seems to mirror the general history of research on NVB in science as described by Paul Ekman (2016)—only with a delay of a couple of decades. Although the role of NVB was already strongly emphasized in the middle of the 19th century (Darwin, 1872), it took nearly 100 years until systematic research began to really take off. With few early exceptions (e.g., Rejeski & Lowe, 1980; Whittaker-Bleuler, 1980), it took further decades until research on NVB in sports began to flourish in the early 2000s. And again, just like it seemed almost trivial by the end of the 1960s that NVB in general conveys meaningful information (Ekman, 2016), with the benefit of hindsight, it now seems nearly as trivial that NVB plays an important role in sports: both as a consequence and as a predictor of behavior, and, maybe most importantly, as a factor influencing the formation of impressions and expectancies, which in turn may impact on sports performance. This, however, should not delude us into thinking that all relevant questions regarding the role of NVB in sports have been answered; quite the contrary is the case. Looking at the research depicted earlier in this chapter and at the open research questions that we have proposed, it becomes clear that we have only begun to uncover the tip of the iceberg. The most urgent open research questions seem to refer to the role of NVB as a predictor and maybe even as a cause of sports performance. While first evidence suggests that NVB can predict performance, this relationship could still be explained by confounding variables and, at present, no direct evidence has shown that NVB can directly influence performance. This might be a very important finding, as it would allow integrating NVB into research on psychological momentum and performance dynamics. In this respect, NVB could prove to be an important mosaic piece in explaining the upward or downward spirals of Matthew effects (Merton, 1968; the rich get richer and the poor get poorer) or Pygmalion effects (Rosenthal & Jacobson, 1968; self-fulfilling prophecies) in sport performance contexts: e.g., when it is going poorly in an athletic contests, this shows in an athlete's NVB, causing opponents to perform better and the athlete to perform worse.

From a practical point of view, incorporating NVB into training and preparation seems fruitful as well. Although NVB plays a role in applied sports psychology, few athletes and coaches probably deal with it as systematically as depicted in the above chapter. Optimally, practical interventions regarding coaches', athletes', and officials' NVB should be theoretically informed and empirically validated to avoid spending resources on ineffective interventions at best and harmful interventions at worst (Gardner & Moore, 2006). Given both the strong research and the strong practical interest into NVB in sports, we are confident that research and practice in this field can reciprocally inform and benefit from each other. In conclusion, acknowledging that Watzlawick's aphorism "one cannot not communicate" holds true in the domain of sports as well (Watzlawick et al., 1967, p. 51), the question remains *how*, *when,* and *what* to communicate. We are confident that the next decade of research and practice on NVB in sports will find answers to these questions.

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